bitop/v1 on a 16–40-bit prime ladder: index calculus with a per-size base

`research/bitop_ladder_20261019` · protocol frozen 2026-10-19 · report 2026-10-20

Question

The bitop/v1 study (PR #1970) fitted exponents with intervals of about ±0.1 from 88 runs an arm on 16–26-bit curves, and ran index calculus at fixed bases only. This study asks:

1. **Precision.** What are the exponents with about 2,000 runs an arm on 25 curves from 16 to 40 bits?
2. **The fair test.** When index calculus chooses its base per curve size on a separate train panel, does its exponent bend toward ½?

Everything here was preregistered in `PROTOCOL.md`, including the verdict rules.

Method

- **Curves.** One `a = −3` prime-order curve per size from 16 to 40 bits (`prime_fast::find_a3_curve`); the 20/24/28/32/36/40-bit ones are the 2026-10-15 campaign's.
- **Phase A, train panel.** Target seed 20261019100: 8 targets per curve up to 34 bits, 4 from 35 to 40 bits.
  - Candidate bases come from a geometric grid with step $2^{\frac{1}{2}}$.
  - The winner is the base with the smallest mean bitop/v1 cost among bases whose train runs all verified and are completely priced.
  - A winner at a grid edge triggers a one-step extension, at most three per size and arm.
  - It took 46 sessions and 2,296 runs: 2,295 verified, and one crashed (the kernel's out-of-memory killer, most likely, in a far-from-optimal candidate).
- **Phase B, hold-out.** Target seed 20261019200, 80 targets a curve, 12,080 runs. Every run verified and is completely priced, and every session passes `ecbench verify --replay 2`.
- **Arms.**
  - rho (negation): the reference;
  - BSGS (negation) and kangaroo: the positive control;
  - an A/A rho;
  - the MITM pair table with the per-size base (16–40 bits);
  - the pair table at the fixed base 32 (16–28 bits);
  - root-finding with $m = 3$ and a per-size base (16–28 bits).
- **Measures.** `ecbench bound fit --unit bitop/v1` (two-stage cluster bootstrap, 2,000 resamples), `ecbench work`, `compare`, `table`, `band` and `amortize`. Counts only, at isolation L0, with one binary (`372ed2eb…`).

Selection (Phase A)

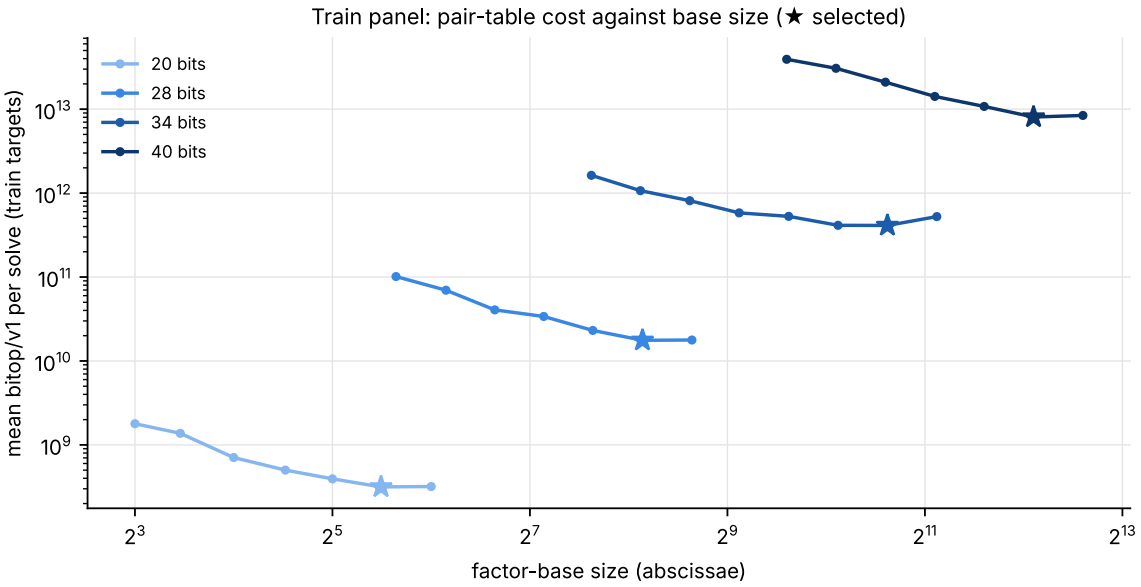


Figure 1: Train panel: mean bitop/v1 per solve against base size, pair table, four sizes (★ selected). Sessions `sessions/train-P*`; values from `select/selection.json`.

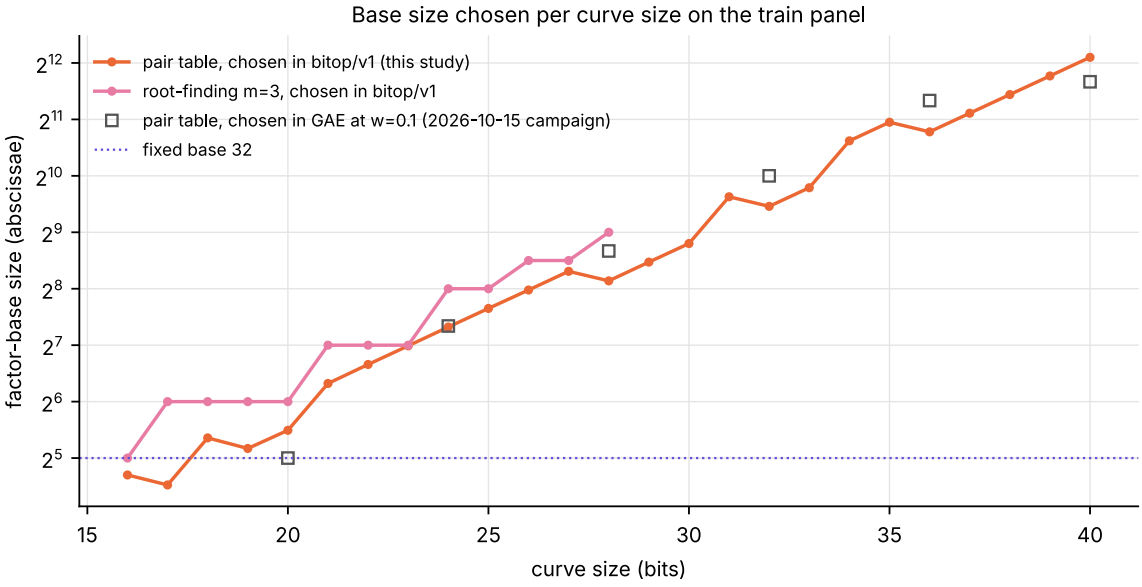


Figure 2: Selected base per size (`params.json`), with the campaign's GAE selection for comparison. The two agree within about a factor of 1.5 in both directions.

Exponents (Phase B)

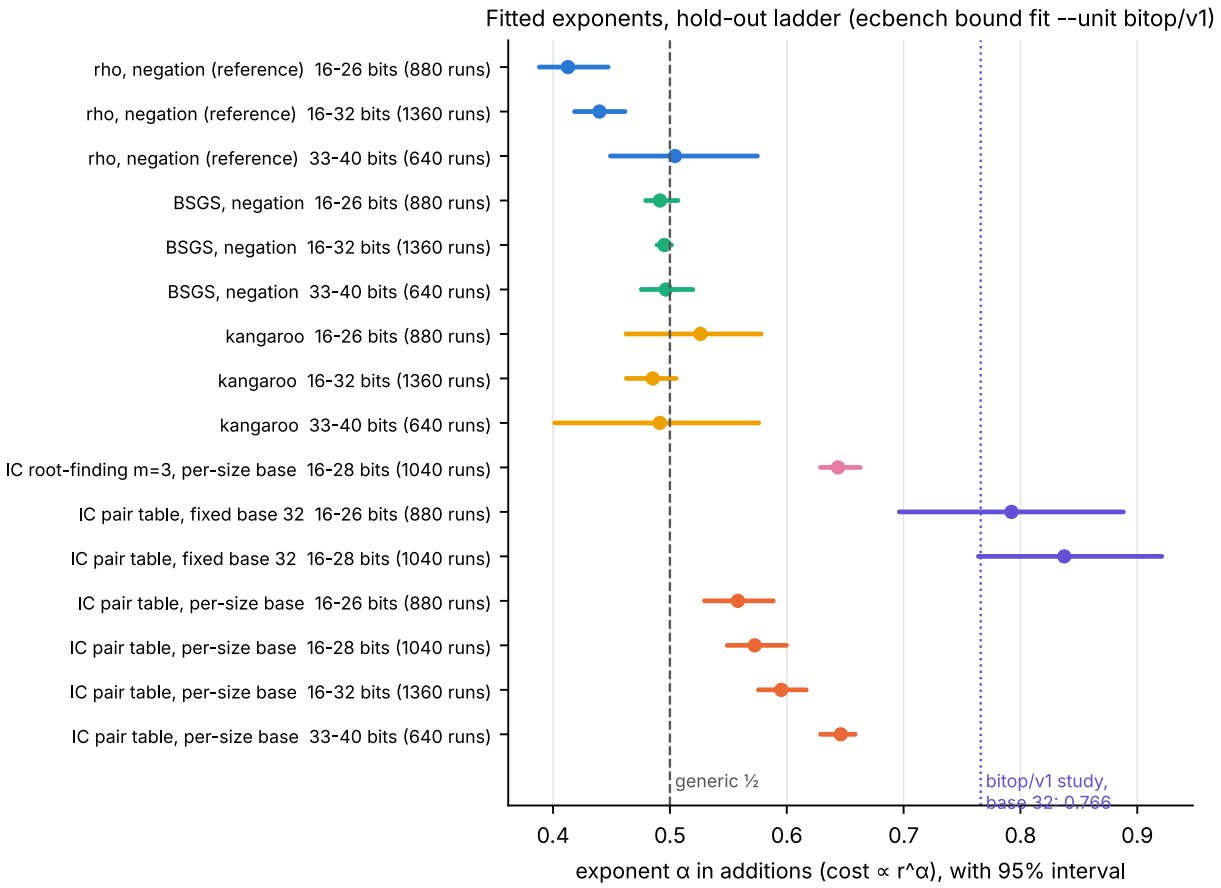


Figure 3: Exponents in additions with 95% intervals, from `analysis/bounds/*.json`. ecbench refuses one fit across its toy (≤ 32 bits) and medium tiers, so the fits are per tier (Deviation 3).

arm	16–26 bits	16–32 bits	33–40 bits
rho, negation	0.413 [0.388, 0.447]	0.440 [0.418, 0.462]	0.504 [0.449, 0.575]
BSGS, negation	0.492 [0.479, 0.507]	0.495 [0.489, 0.501]	0.497 [0.475, 0.520]
kangaroo	0.526 [0.462, 0.578]	0.485 [0.463, 0.505]	0.491 [0.401, 0.576]
IC pair, per-size base	0.558 [0.530, 0.588]	0.595 [0.576, 0.617]	0.646 [0.629, 0.659]
IC pair, base 32	0.792 [0.696, 0.888]	16–28: 0.838 [0.764, 0.921]	—
IC root-finding, per-size	—	16–28: 0.644 [0.629, 0.663]	—

Precision. Interval widths at 16–32 bits:

- BSGS: 0.012;
- the IC pair table: 0.041;
- rho: 0.043.

PR #1970's 16–26-bit ladder had widths of 0.07–0.31.

Positive control. It trips once. Rho's 16–26-bit interval excludes ½ by 0.053, above the protocol's 0.05 tolerance.

- **Other ranges:** rho at 16–32 bits misses ½ by 0.038, within the tolerance, and rho at 33–40 bits contains ½.
- **BSGS:** contains ½ in every range.
- **Pattern:** rho's constant falls with size (1.47, 1.33 and 1.05 times the floor). That is consistent with fixed per-walk overhead on small curves. This explanation was not tested.

A/A control. Uninformative: it reuses rho's seeds, so its runs are identical copies (Deviation 4).

Distance to rho

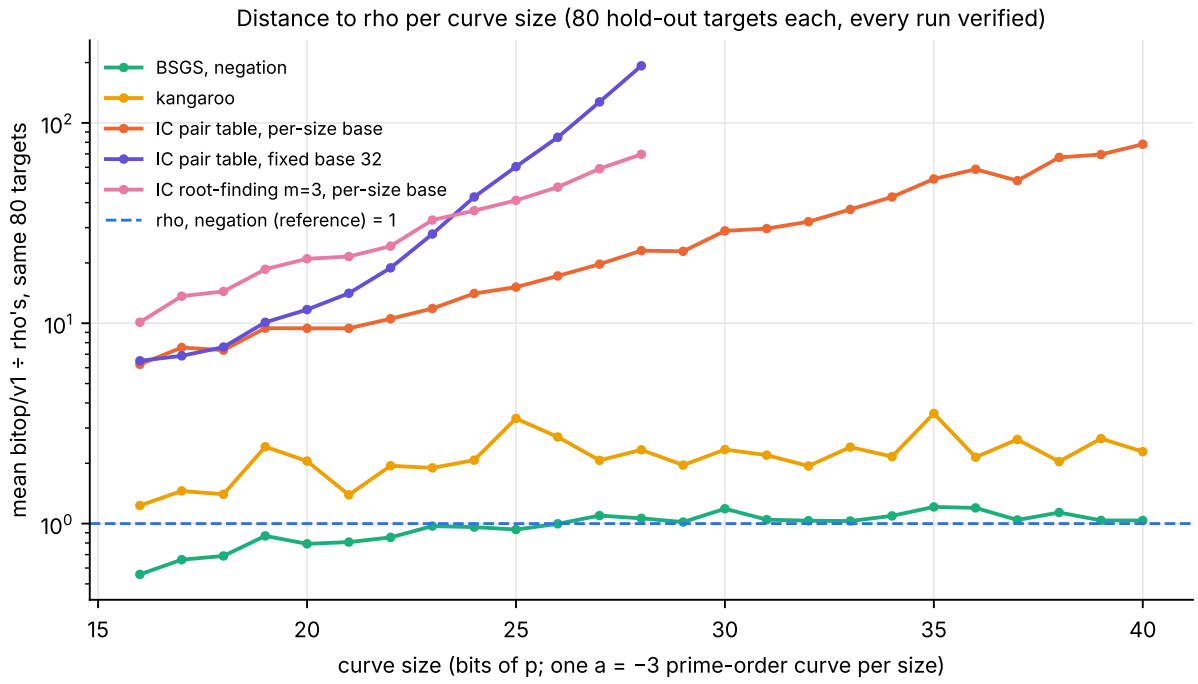


Figure 4: Mean bitop/v1 over rho's, per curve, 80 hold-out targets each (`analysis/gap_to_rho.json` from `ecbench work --counted`).

Verdicts (preregistered)

- **Step 2: bends toward ½.**
  - The per-size pair table's 16–26-bit upper bound, 0.588, is below the bitop/v1 study's fixed-base 0.766.
  - Its 16–28-bit interval, [0.549, 0.600], lies below the fixed base's from this study, [0.764, 0.921].
- **Success criterion: met.** This is a verified, measured improvement of the index-calculus baseline in bitop/v1. The fixed base costs 1.0×, 3.0× and 8.4× the per-size base at 16, 24 and 28 bits.
  - The §5a band orders the two at every size from 18 to 28 bits, at all three weights.
  - At 17 bits the fixed base is cheaper; at 16 bits they are not separated.
- **Distance remaining.** Rho is cheaper at every size and band weight: the per-size pair table costs 6.2× rho at 16 bits, 32× at 32 bits and 78× at 40 bits. No curve has an amortization window (0 of 25).
- **Trend.** The pair table's exponent rises with size: 0.558, then 0.595, then 0.646, with separated intervals. Index calculus moves away from rho as the curves grow.

Scope and limits

- **Curves and claims.** Public synthetic curves and targets; no security claim.
- **Counts only.** No wall time is used.
- **Curve sample.** One curve per size.
- **Deviations.** Four are recorded in `PROTOCOL.md`: the cards were split by the agent harness's time limit, the extension files were renamed, the fits are per tier, and the A/A control is degenerate.
- **Provenance.** The recorded git commit varies between sessions, because documentation commits landed during runs. The binary is identical throughout.